

Manish Reddy Pannala

Data Engineer | Snowflake | GCP | AWS | Python | ETL Pipelines

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SUMMARY

Data Engineer with 2+ years of experience designing and operationalizing data pipelines, models, and integrations across cloud platforms (Snowflake, GCP, AWS). Skilled in building CI/CD-driven ETL frameworks, improving data quality, and enabling advanced analytics and machine learning initiatives. Experienced with Python, SQL, API integrations, and Power BI to deliver scalable, reliable enterprise data solutions.

EDUCATION

Drexel University - MSc Data Science, Conc: Data Engineer

Jan 2024 – Sep 2025

VIT University – B. Tech Electronics and Communication Engineering

Jun 2017 – May 2021

PROFESSIONAL EXPERIENCE

Research Assistant @Drexel University, Philadelphia, PA

Sep 2024 – Jun 2025

- Built **NLP pipelines** using **Python**, **BERT**, and **GPT** to automate text classification, summarization, and policy rule-violation detection across social datasets.
- Designed **data infrastructure** for large-scale social media datasets, enabling efficient storage, retrieval, and downstream model experimentation.
- Developed scripts to identify & remove advertisement segments from podcast transcripts by detecting repeated sentence patterns.
- Fine-tuned transformer models to improve text extraction accuracy and research workflow automation.

Data Engineer @RandomTrees Pvt Ltd – Hyderabad, India.

Sep 2022 – Dec 2023

- Executed machine learning models within **Snowflake** using Snowpark, delivering multiple proof-of-concept (POC) projects for internal teams and products. Conducted POCs on **Snowflake**, **AWS** and **GCP** (individually and in combination), enhancing cloud integration capabilities across the organization.
- Explored and implemented **GCP services (Compute Engine, Cloud Storage, API Management)** to automate tasks, enable efficient storage, and support cloud migration.
- Developed comprehensive technical documentation to support cross-team communication and workflow clarity.

DFast Accelerator (Database Migration Tool)

- Developed DFast, a data migration suite enabling seamless migration from legacy systems (**Oracle**, **MySQL**) to **Snowflake**.
- Implemented Python-based extract-transform-load automation & backend APIs. Implemented secure **authentication** and authorization features for user management.
- Enabled faster enterprise cloud onboarding and reduced manual SQL rewrite effort.
- Built **FastAPI + SQLAlchemy APIs** for migration processes, integrated audit/log tables, and configured **PostgreSQL** as backend infrastructure.

MicroStrategy Sunset Migration Project

- Migrated enterprise reporting systems from Oracle & MicroStrategy to Snowflake & Power BI for 50+ reports.
- Designed Matillion ETL pipelines for daily, weekly, and monthly data ingestion from APIs and databases into Snowflake.
- Automated post-migration validation using Python-based hash checks, ensuring 100% data accuracy.
- Optimized SQL queries, reducing report load times by 40%.
- Authored documentation and led troubleshooting handoffs for long-term maintenance.

PROJECTS

Bank Risk Prediction & Financial Dashboard — Python, SQL, Power BI, FDIC Data

- Built financial distress prediction model and executive dashboard for bank risk assessment.
- Automated FDIC data ingestion and developed KPIs for capital strength, liquidity, and profitability.
- Delivered drill-down reporting with early-warning insights for high-risk institutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Cloud: Google Cloud Platform, AWS(S3,EC2)

Data Warehouses: Snowflake, BigQuery, Oracle, PostgreSQL

ETL & Orchestration: Matillion, Airflow, Snowpark, FastAPI, GitLab CI/CD

Tools: Git, Power BI, Linux, Docker

Concepts: Data Modeling, ELT/ETL, API Development, Cloud Migration, SQL Optimization